

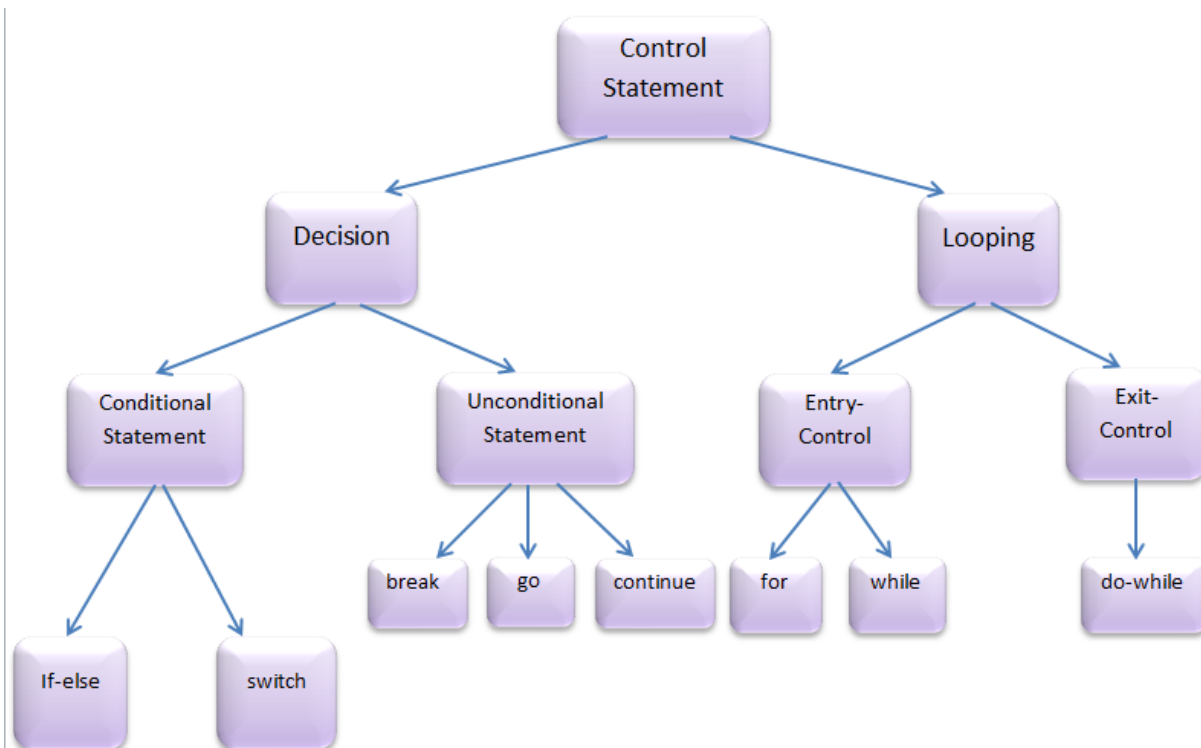
# Control Statements

## What is Control Statement?

**Control statements** enable us to specify the flow of program control; such as the order in which the instructions in a program must be executed. They make it possible to make decisions, to perform tasks repeatedly or to jump from one section of code to another.

There are four types of control statements in C:

1. Decision making statements
2. Selection statements
3. Iteration statements
4. Jump statements



## Switch

Switch statement in C tests the value of a variable and compares it with multiple cases. The value provided by the user is compared with all the cases inside the switch block until the match is found. If a case match is NOT

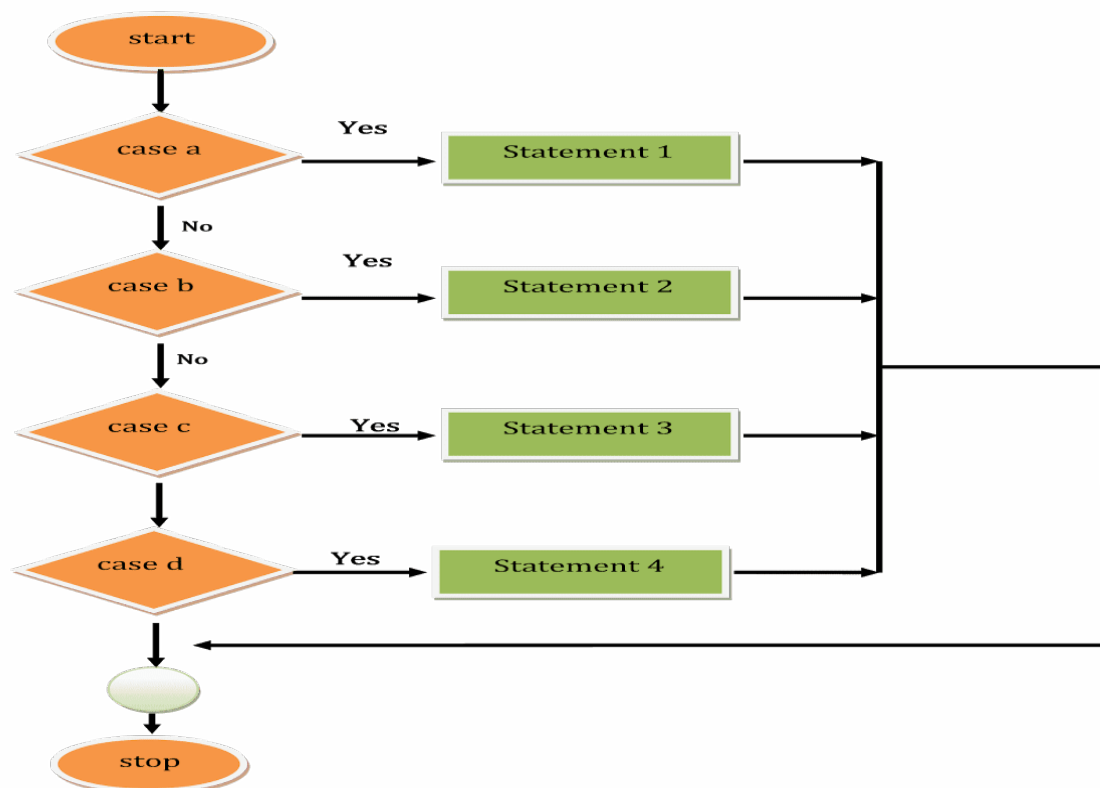
found, then the default statement is executed, and the control goes out of the switch block. Switch case statement checks the integer or character value in multiple case statements.

### Syntax

Switch (expression)

```
{  
case 1: Statement 1;  
case 2: Statement 2;  
case 3: Statement 3;  
default;  
}
```

### Switch case flow chart



### How does a switch statements work?

First switch statement examines condition then it will find the corresponding case. If the condition is not fulfilled then the compiler moves to further cases until the condition is not completed.

**Example: Check a number even or odd**

```

{
int n;
printf("Choose a number\n");
scanf("%d",&n);
switch(n%2)
{
case 0: printf("It's Even Number");
break;
case 1: printf("It's Odd Number");
}
}

```

Output: Choose a number

6

It's Even Number

**Example: Find the name of a day.**

```

{
int week;
printf("Enter week number (1-7): ");
scanf("%d", &week);
switch(week)
{
case 1:printf("Monday");
break;
case 2: printf("Tuesday");
break;
case 3: printf("Wednesday");
break;
case 4: printf("Thursday");
break;
case 5: printf("Friday");
break;
case 6: printf("Saturday");
break;
case 7: printf("Sunday");
break;
default: printf("Invalid input! Please enter week number between 1-7.");
}
return 0;
}

```

**Output :**

**Input**

Enter week number (1-7): 2

Tuesday

### Why we use break statements with switch cases?

We use break statement with cases. The break statement takes cursor out from the switch block. If we do not use break statement then compiler will check for other conditions which will not be satisfied. Also, it will increase the execution time.

### Comparison between if-else and switch

| Basis For Comparison  | IF-ELSE                                                                                                    | SWITCH                                                                                                                                        |
|-----------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Basic                 | Which statement will be executed depends upon the output of the expression inside if statement.            | Which statement will be executed is decided by user.                                                                                          |
| Expression            | If-else statement uses multiple statements for multiple choices.                                           | Switch statement uses single expression for multiple choices.                                                                                 |
| Testing               | If-else statement tests for equality as well as for logical expression.                                    | Switch statement tests only for equality.                                                                                                     |
| Evaluation            | If statement evaluates integer, character, pointer or floating-point type or boolean type.                 | Switch statement evaluates only character or integer value.                                                                                   |
| Sequence of Execution | Either if statement will be executed or else statement is executed.                                        | Switch statement executes one case after another till a break statement is appeared or the end of switch statement is reached.                |
| Default Execution     | If the condition inside if statements is false, then by default the else statement is executed if created. | If the condition inside switch statements does not match with any of cases, for that instance the default statements are executed if created. |
| Editing               | It is difficult to edit the if-else statement, if the nested if-else statement is used.                    | It is easy to edit switch cases as they are recognized easily.                                                                                |

### Practice Examples

1. Check whether an alphabet is vowel or consonant using switch case
2. Print number of days in a month using switch-case
3. Find maximum between two numbers using switch case
4. Check whether a number is positive, negative or zero using switch case.
5. Program to find roots of a quadratic equation using switch case.
6. Create Simple calculator using switch case.